

EXERCISE-5

1. Filter unique array members using Set.

```
JS q1.js > ...  
1  const arr =[1,2,2,3,4,4,5,6,6];  
2  const uniqueMem=[... new Set(arr)];  
3  console.log(uniqueMem);
```

Output:

```
[Running] node "/home/harsh-raj/Bootcamp_Assignment/js_exercise5/q1.js"  
[ 1, 2, 3, 4, 5, 6 ]
```

2. Filter anagrams using Map.

```
JS q2.js > [⌕] filter_anagram
1  let filter_anagram=(str1,str2)=>{
2    if(str1.length!==str2.length){
3      return false;
4    }
5
6    let map = new Map();
7    str1=str1.toLowerCase();
8    str2=str2.toLowerCase();
9
10   for(let c of str1){
11     map.set(c, (map.get(c)||0)+1);
12   }
13
14   for(let c of str2){
15     if(!map.has(c))
16       return false;
17     map.set(c,map.get(c)-1);
18     if(map.get(c)<0)
19       return false;
20   }
21   return true;
22 };
23
24 console.log(filter_anagram("ehreho","heeroh"))
25 console.log(filter_anagram("Harsho","Yogesh"));
26
```

Output:

```
[Running] node "/home/harsh-raj/Bootcamp_Assignment/js_exercise5/q2.js"
true
false
```

Tested the code for both true and false values.

3. Write a program to implement inheritance upto 3 classes. The Class must contain private and public variables and static functions.

```
js q3.js > Customer > showCustomer
1  class Registration {
2    #registration_id=0;
3    constructor(registration_id){
4      this.#registration_id=registration_id;
5    }
6
7    static showRegistration(obj){
8      console.log("Registration-Id: ", obj.#registration_id);
9    }
10 }
11
12 class Customer extends Registration{
13   #age=0;
14   constructor(registration_id,name,age){
15     super(registration_id);
16     this.name=name;
17     this.#age=age;
18   }
19
20   static showCustomer(obj){
21     this.showRegistration(obj);
22     console.log("Customer-Name: ", obj.name);
23     console.log("Customer-Age: ", obj.#age);
24   }
25 }
26
27 class Car extends Customer{
28   #id=0;
29   constructor(registration_id,name,age,modelName,id){
30     super(registration_id,name,age);
31     this.modelName=modelName;
32     this.#id=id;
33   }
34
35   static showCar(obj){
36     this.showCustomer(obj);
37     console.log("Model-Name: ", obj.modelName);
38     console.log("Id: ", obj.#id);
39   }
40 }
41
42 const carSale=new Car(101,'Harsh',21,'Chevrolet-CX',1001);
43 Car.showCar(carSale);
44
```

Output:

```
[Running] node "/home/harsh-raj/Bootcamp_Assignment/js_exercise5/q3.js"
Registration-Id: 101
Customer-Name: Harsh
Customer-Age: 21
Model-Name: Chevrolet-CX
Id: 1001
```

4. Write a program to implement a class having static functions

```
JS q4.js > StudentInstance
1  class Student{
2      constructor(name,age,student_id){
3          this.name=name;
4          this.age=age;
5          this.student_id=student_id;
6      }
7
8      static showDetails(name,age,student_id){
9          console.log("Name: ", name);
10         console.log("Age: ", age);
11         console.log("Student: ", student_id);
12     }
13 }
14
15
16 Student.showDetails('Harsh',21,101);
17
18 const StudentInstance=new Student({});
19 try{
20     StudentInstance.showDetails('Harsh',21,101);
21 }catch(e){
22     console.log("Instance cannot call static method: ", e.message);
23 }
```

Output:

```
[Running] node "/home/harsh-raj/Bootcamp_Assignment/js_exercise5/q4.js"
Name: Harsh
Age: 21
Student: 101
Instance cannot call static method: StudentInstance.showDetails is not a function
```

5. Import a module containing the constants and method for calculating area of circle, rectangle, cylinder.

```
JS q5.js > ...
1 import { areaOfCircle,areaOfCylinder,areaOfRectangle,PI} from "./q5_area.js";
2
3 console.log(PI);
4 console.log(areaOfCircle(5));
5 console.log(areaOfRectangle(3,6));
6 let [curvedSurfacearea,totalSurfaceArea]=areaOfCylinder(5,12);
7
8 console.log(`The Curved Surface area of the Cylinder is ${curvedSurfacearea}`);
9 console.log(`The Total Surface area of Cylinder is ${totalSurfaceArea}`);
```

Module:

```
JS q5_area.js > areaOfRectangle
1 export const PI=3.14;
2
3 export function areaOfCircle(radius){
4   let area=PI*Math.pow(radius,2)
5   return area;
6 }
7
8 export function areaOfRectangle(length,breadth){
9   let area = length*breadth;
10  return area;
11 }
12
13 export function areaOfCylinder(radius,height){
14   let curvedSurfaceArea=2*PI*radius*height;
15   let totalSurfaceArea=curvedSurfaceArea+(2*PI*Math.pow(radius,2));
16   return [curvedSurfaceArea,totalSurfaceArea];
17 }
```

Output:

3.14	q5.js:3
78.5	q5.js:4
18	q5.js:5
The Curved Surface area of the Cylinder is 376.8	q5.js:8
The Total Surface area of Cylinder is 533.8	q5.js:9

6. Import a module for filtering unique elements in an array.

```
JS q6.js
1 import { filterWithSet,filterWithoutSet } from "./q6_filter.js";
2
3 console.log(`Filter with set ${filterWithSet([1,2,5,2,4,7,4,8])}`);
4
5 console.log(`Filter without set ${filterWithoutSet([1,2,4,5,8,3,5,7,2,4])}`);
```

Module:

```
JS q6_filter.js > filterWithSet
1 export function filterWithSet(arr){
2   let s=new Set();
3   arr.forEach(element => {
4     s.add(element);
5   });
6
7   return [...s];
8 }
9
10 export function filterWithoutSet(arr){
11   return arr.filter((e,i,self)=> i===self.indexOf(e));
12 }
```

Output:

Filter with set 1,2,5,4,7,8	q6.js:3
Filter without set 1,2,4,5,8,3,7	q6.js:5

7. Write a program to flatten a nested array to single level using arrow functions.

```
JS q7.js > ...  
1  const arr=[1,2,[3,4,5,[6,7]]];  
2  const flattenNested = arr => arr.flat(Infinity);  
3  console.log(flattenNested(arr));|
```

Output:

```
[Running] node "/home/harsh-raj/Bootcamp_Assignment/js_exercise5/q7.js"  
[  
  1, 2, 3, 4,  
  5, 6, 7  
]
```